

Lecture # 1

Wednesday
6 March 2024
9-10 AM
4-5 PM

PH102 After Mid Semester Exams

by

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The Lorentz Force Law, Biot-Savart Law **Chapter 5**

[“Introduction to Electrodynamics” by David J. Griffiths](#)

General Introduction

Historical overview:

Chinese (13th century BC) used a compass needle that it was responding to the Earth's magnetic field.

Greeks (8th century BC) discovered certain stones, called magnets, generally made by Magnetite iron(II,III) oxide = Fe_3O_4



Magnetite



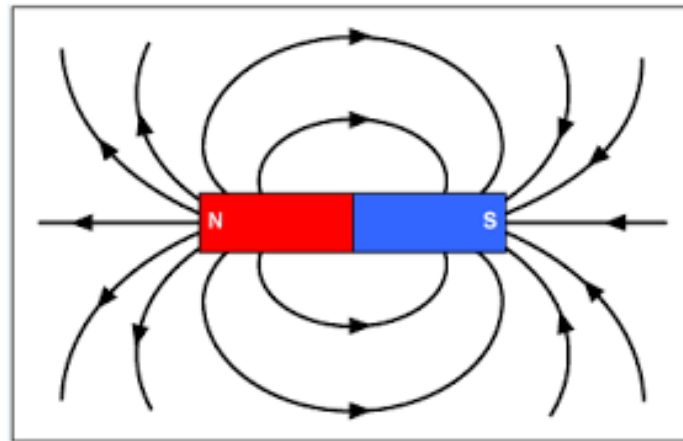
Magnetic Poles: Every magnet has a north and a south pole which produce a magnetic force on each other.

Source of magnetism



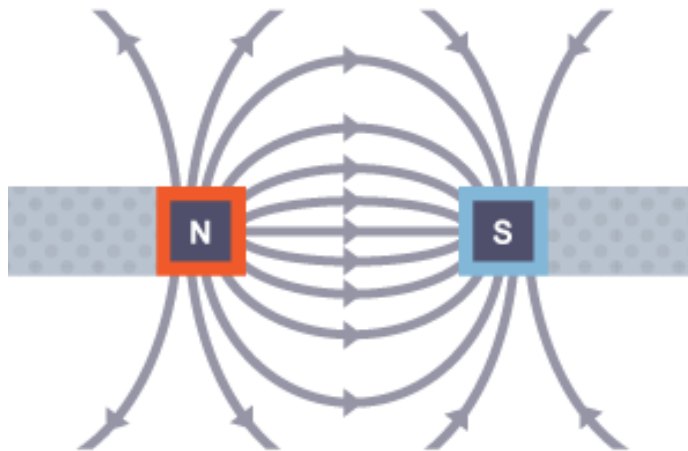
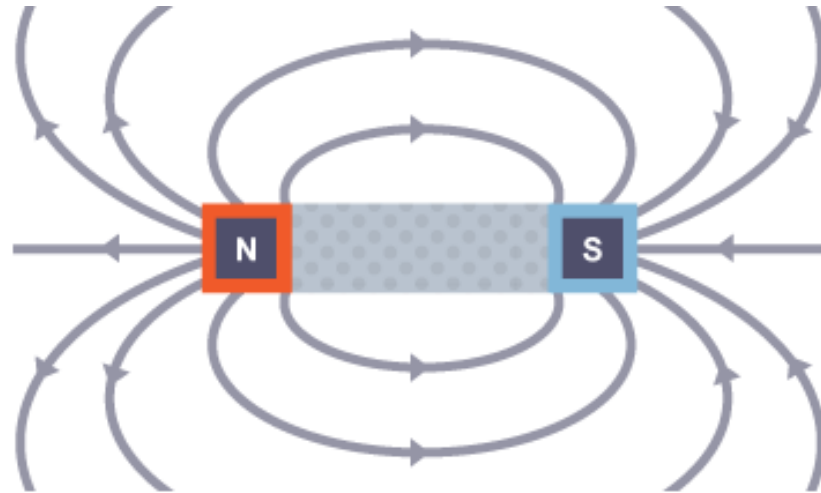
Electric Charge

The electric force or field is generated by electric charge and the magnetic force or field by charge in motion.

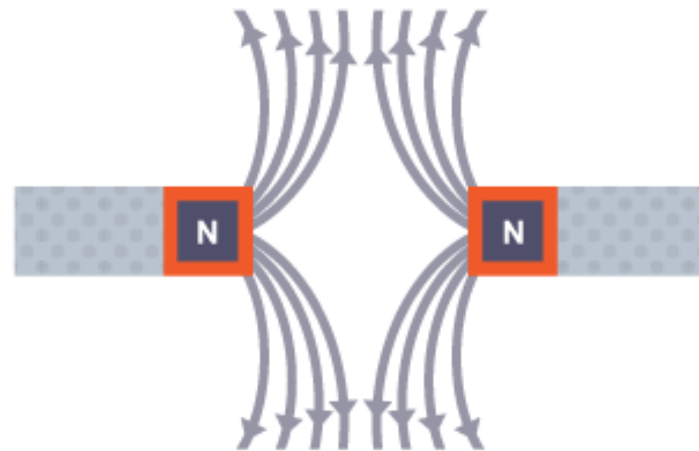


Magnetic field lines: The magnetic field of a bar magnet can be traced with the aid of a compass, defining magnetic field lines, similar in many ways to the electric field lines. $\mathbf{B}$ is tangent on the magnetic field lines, with north-to-south direction. The density of the lines is proportional to the strength of $\mathbf{B}$.

General Introduction

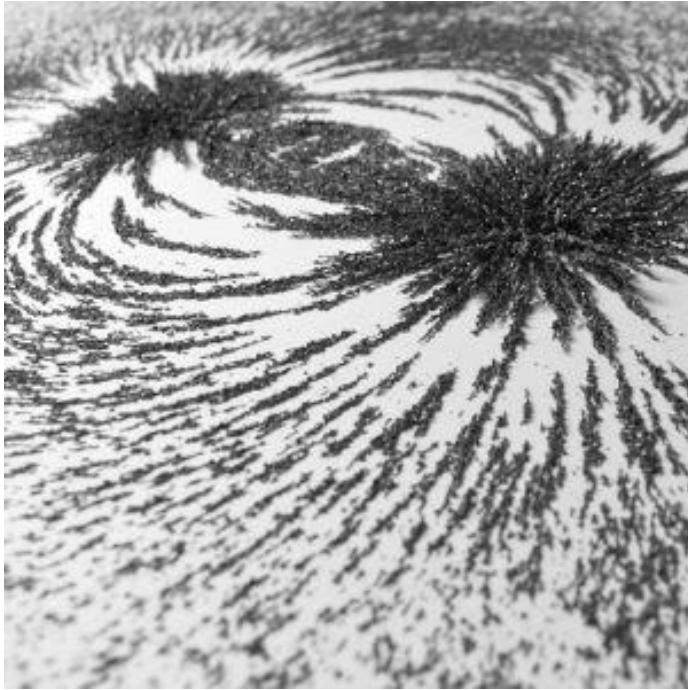


Unlike pole attraction



Like pole repulsion

Magnetic field lines can be displayed by small iron fillings placed in the vicinity of any shape magnet.



$$B_{\text{earth}} \sim 0.65 \text{ Oe (or Gauss)}$$

$$B_{\text{sun}} \sim 1 \text{ Oe (or Gauss)}$$

$$B_{\text{sun-spots}} \sim 4000 \text{ Oe (or Gauss)}$$

Magnetic Flux Density $\mathbf{B} = \Phi_{\mathbf{B}}$ (Magnetic Flux) per Unit Area

$$\text{Units} = 1 \text{ Web/m}^2 \text{ (or Tesla or (N. S)/(C. m))}$$

Units of B in CGS units Oersted (Oe)

$$\text{Tesla} = 10000 \text{ Oe}$$

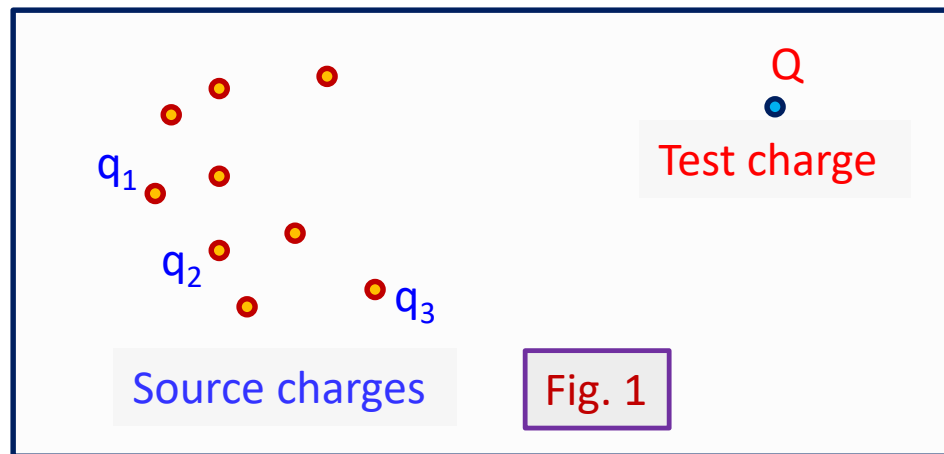
The magnetic flux, represented by the symbol Φ , threading some contour or loop is defined as the magnetic field $\mathbf{B}$ multiplied by the loop area $\mathbf{S}$, i.e. $\Phi = \mathbf{B} \cdot \mathbf{S}$. Obviously, both $\mathbf{B}$ and $\mathbf{S}$ can be arbitrary and so is Φ .

Magnetic flux quantum $\Phi_0 = h/(2e)$

$\Phi_0 = h/(2e) \approx 2.067833758(46) \times 10^{-15} \text{ Wb}$ is a combination of fundamental physical constants: the Planck constant h and the electron charge e .

The Lorentz Force Law

- Up to now we have restricted our attention to the simplest case of electrostatics, where the charges, q_1 , q_2 , q_3 etc. (source charges) are stationary and the force exerted by these charges on another charge Q (test charge) was determined by superposition principle. Here, it is sufficient to determine the force of a single source charge and the total force is the vector sum of all individual forces.
- Now, we are going to consider the forces between the charges, when they are in motion.



- To understand the forces involved, when charges are in motion, consider two parallel wires having current I flowing through the system as shown in Fig.2. The moment, battery / power supply is switched on, the wires jump apart. If we keep a test charge Q close to one of the wire, you don't see any force on the test charge. So, the force between the wires is not due to the electrical repulsion of like charges or in general electrostatic forces.
- On the other hand, if the parallel parts of the wires are arranged such that current direction is parallel to each other, the wires attract each other.

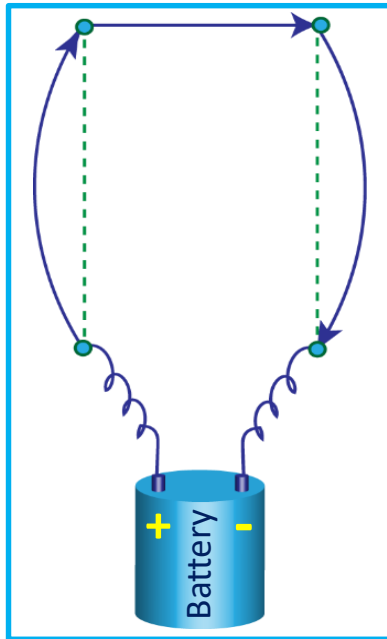


Fig: 2a

The force is basically due to magnetic field produced by moving charges ?

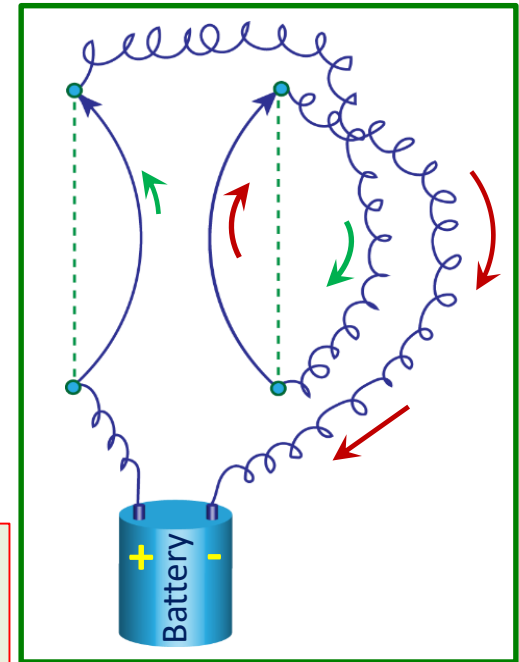
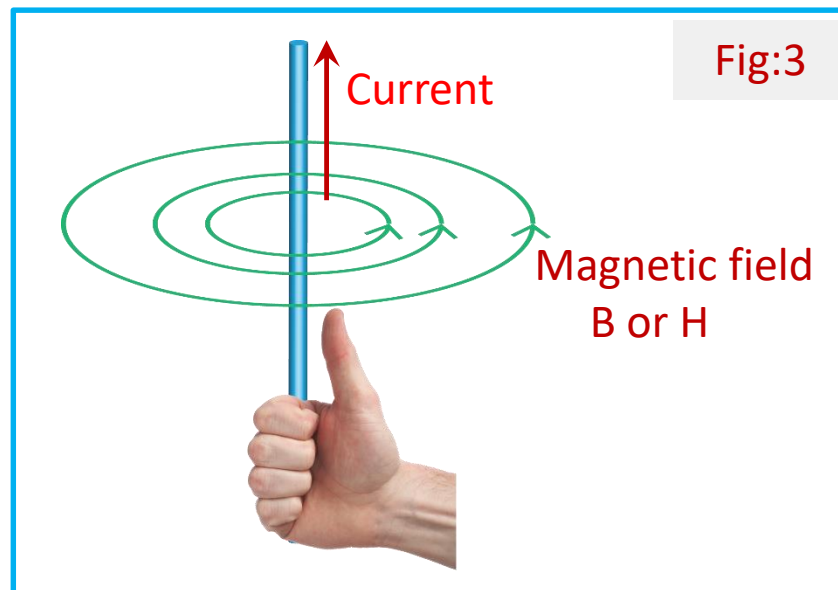


Fig: 2b

- The force is basically due to **magnetic field** produced by moving charges and can be seen by placing a tiny magnetic **compass** close to the current carrying wires.
- The pointing of the north pole of the magnetic compass needle shows the direction of the magnetic field. We can find that the direction of magnetic field is neither along or away from the current carrying straight wire but it **circles around** the wires.
- Grabbing the wire with **right hand with thumb** pointing along the direction of current, the **curled fingers** around the wire represent the direction of magnetic field as shown in Fig. 3.



- How such a field, $\mathbf{B}$ leads to force of attraction between parallel current carrying wires? The direction of $\mathbf{B}$ (due to first wire) is into the page and the velocity of charges (positive) are upwards and yet the force of attraction is towards left as shown in below Fig.
- Lorentz law explains such Force direction.

Attraction Force between parallel current carrying wires

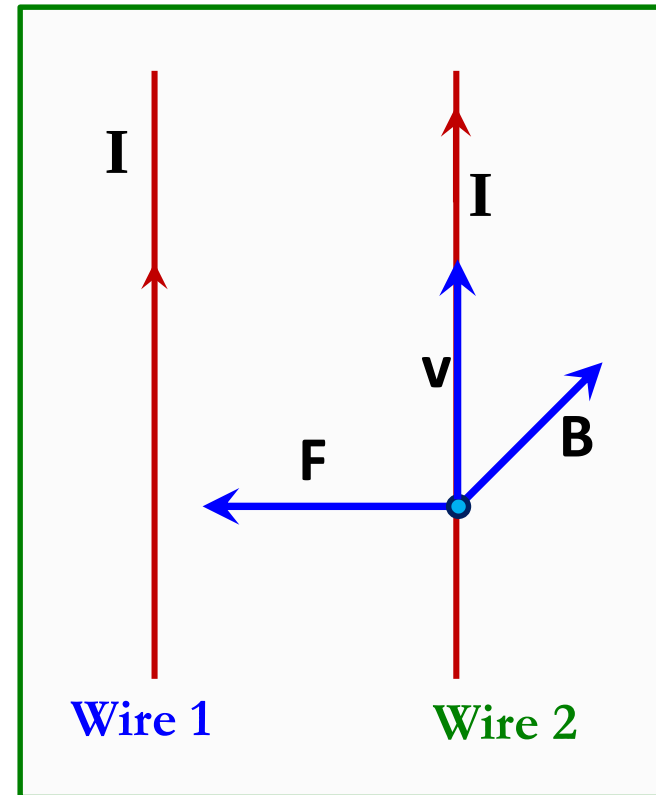
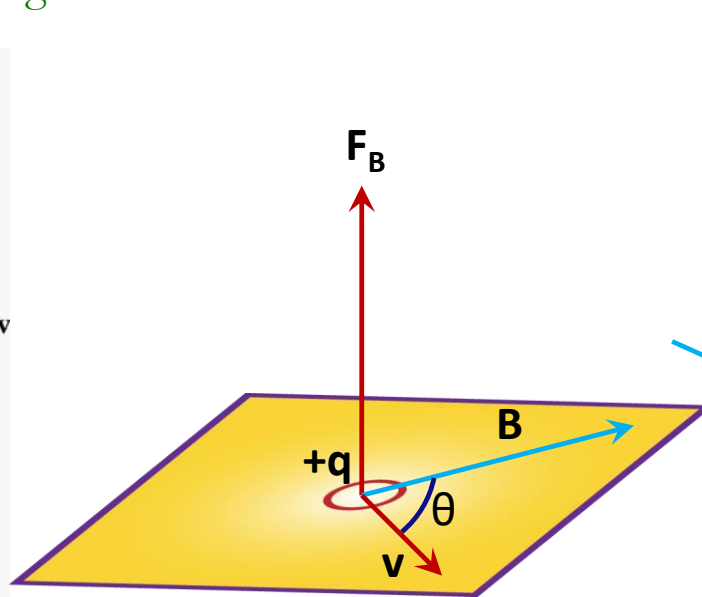
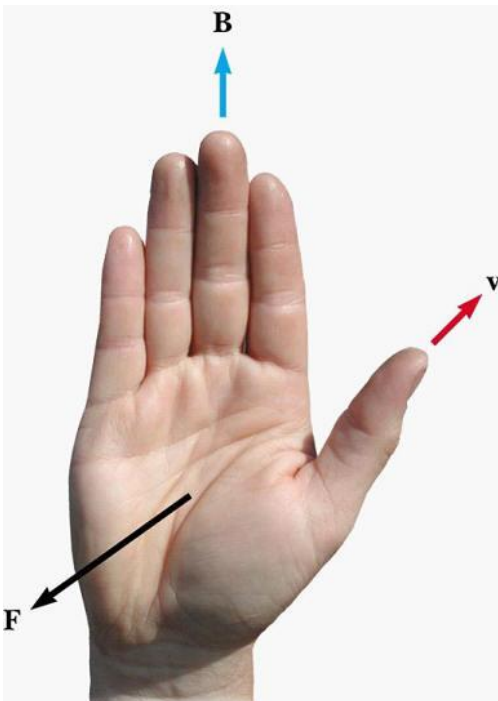


Fig.4

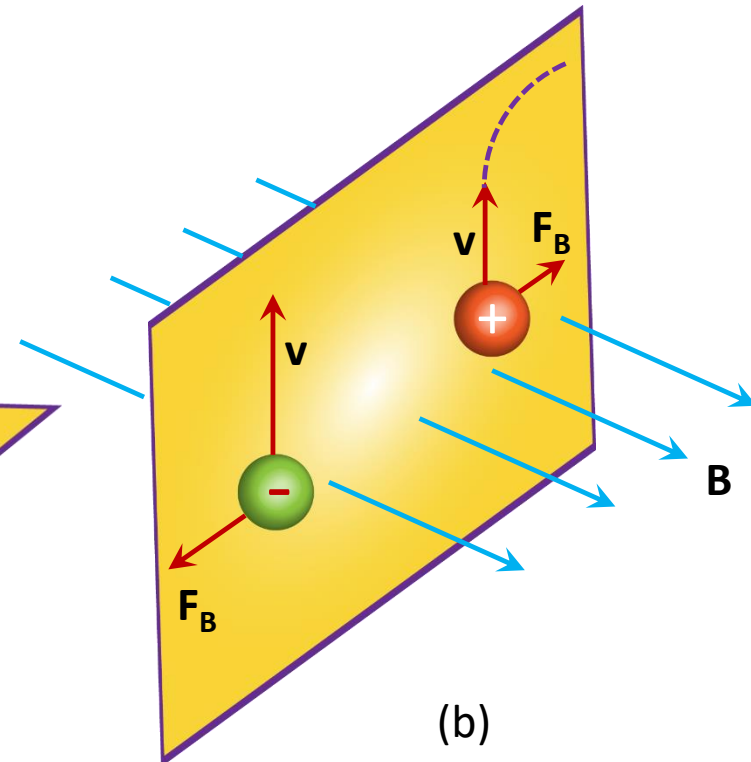
Magnetic force and magnetic field – Quantitatively

Observations: The existence of $\mathbf{B}$ somewhere in space can be determined by measuring the magnetic force $\mathbf{F}$ exerted on a moving charge. Such an experiment finds:

- $\mathbf{F}$ is proportional on q , $\mathbf{v}$ and $\mathbf{B}$.
- $\mathbf{F}$ is max when $\mathbf{v}$ is perpendicular to $\mathbf{B}$ and zero when $\mathbf{v}$ is parallel to $\mathbf{B}$.
- $\mathbf{F}$ is perpendicular to both $\mathbf{v}$ and $\mathbf{B}$. The direction of the force on a positive charge is found with the right hand rule. The direction of the force $\mathbf{F}$ on a negative charge is opposite to the force on a positive charge moving in the same direction.



(a)

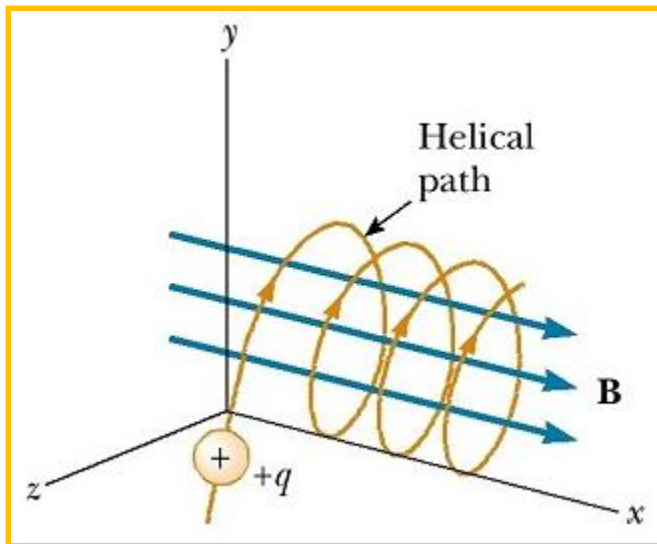


(b)

Lorentz-law: Fundamental axiom of the theory whose justification is to be found in experiments such as the one we saw in earlier slide.

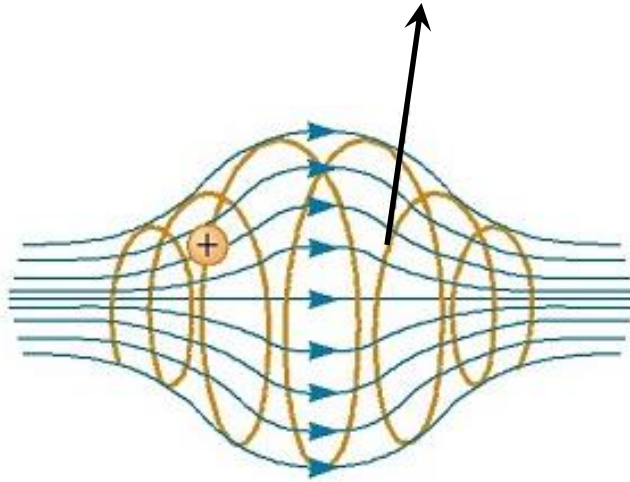
Our main job from now on is to **calculate the magnetic field $\mathbf{B}$** .

But before we proceed, it is worthwhile to take a closer look at the Lorentz force law itself; it is a peculiar law, and it leads to some truly **bizarre particle trajectories**.

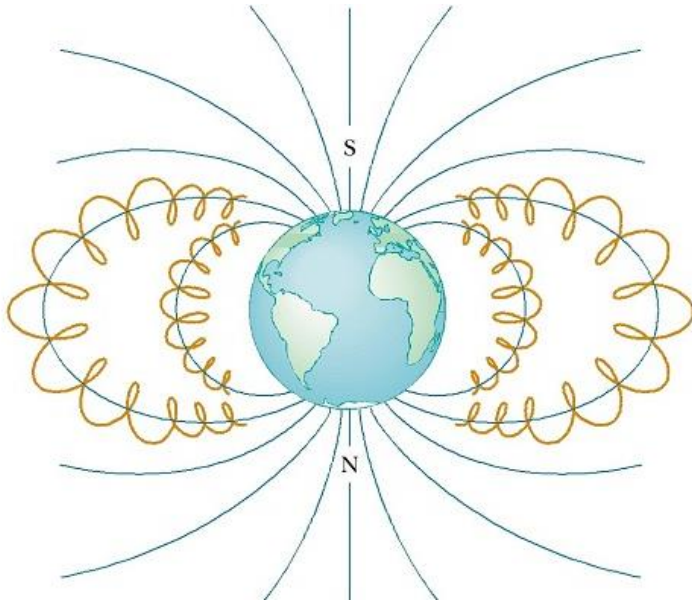


A charged particle having a velocity vector that has a component parallel to the uniform magnetic field moves in helical path

Path of Particle



A charged particle moving in a **non-uniform magnetic field** (**Magnetic Bottle**) spirals about the field (yellow path) and oscillates between the end points. The magnetic force exerted on the particle near either end of the bottle has a component that causes the particle to spiral back towards the center.



The Van Allen belts are made up of charged particles trapped by the Earth's **non-uniform magnetic field**. The magnetic fields line are blue and the particle paths in dark yellow.

Magnetic Forces:

- The magnetic force in a charge Q moving with a velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ is,

$$\mathbf{F}_{mag} = Q(\mathbf{v} \times \mathbf{B}) \quad \text{--- (1.1)}$$

- This is known as **Lorentz** force law. This explains the orthogonal direction of $\mathbf{v}$, $\mathbf{B}$ and $\mathbf{F}$ as shown in Fig 4.
- In the presence of both electric and magnetic fields, the net force on Q would be,

$$\mathbf{F} = Q[\mathbf{E} + (\mathbf{v} \times \mathbf{B})] \quad \text{--- (1.2)}$$

- We have not derived these equations, the justification of these equations can be found from the experimental result.

- One important feature of magnetic force law is that **magnetic forces do no work**.
- If Q moves an amount $d\mathbf{l} = \mathbf{v} dt$, the work done is,

$$dW_{mag} = \mathbf{F}_{mag} \cdot d\mathbf{l} = Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = 0 \quad \text{--- (1.3)}$$

This is because $(\mathbf{v} \times \mathbf{B})$ is perpendicular to $\mathbf{v}$.

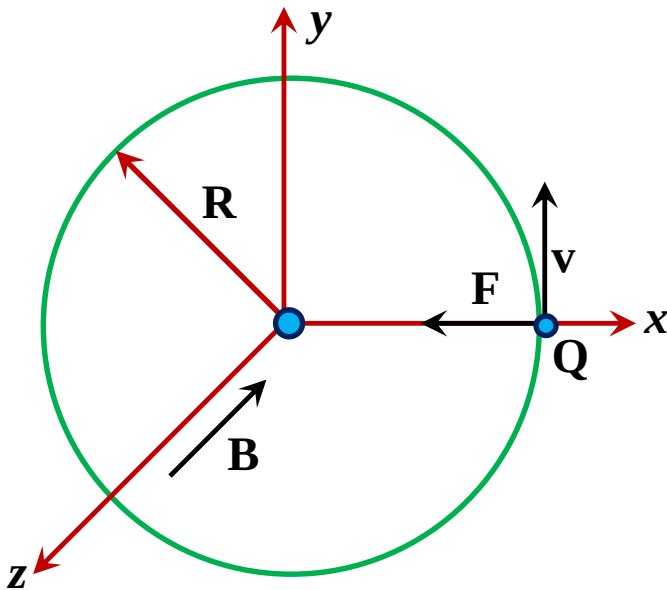
- The fact that the magnetic forces do no work is an elementary and the direct consequence of Lorentz force law.
- We can understand this fact by taking several examples.

Cyclotron motion

The archetypical motion of a charged particle in a magnetic field is circular, with the magnetic force providing the centripetal acceleration.

In below figure, a uniform magnetic field points into the page; if the charge Q moves counterclockwise, with speed v , around a circle of radius R , the magnetic force points inward, and has a fixed magnitude QvB (Just right to sustain uniform circular motion :)

“Cyclotron formula” one can measure the radius of its circular trajectory.



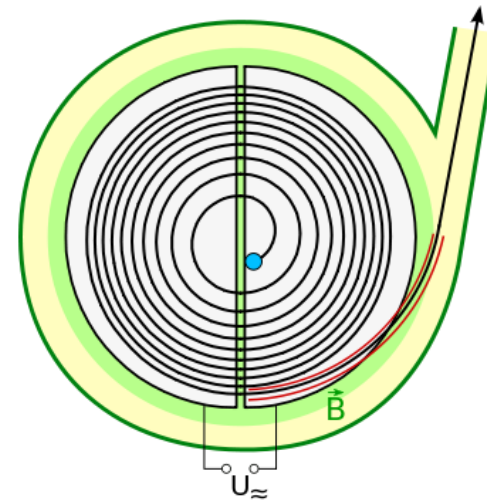
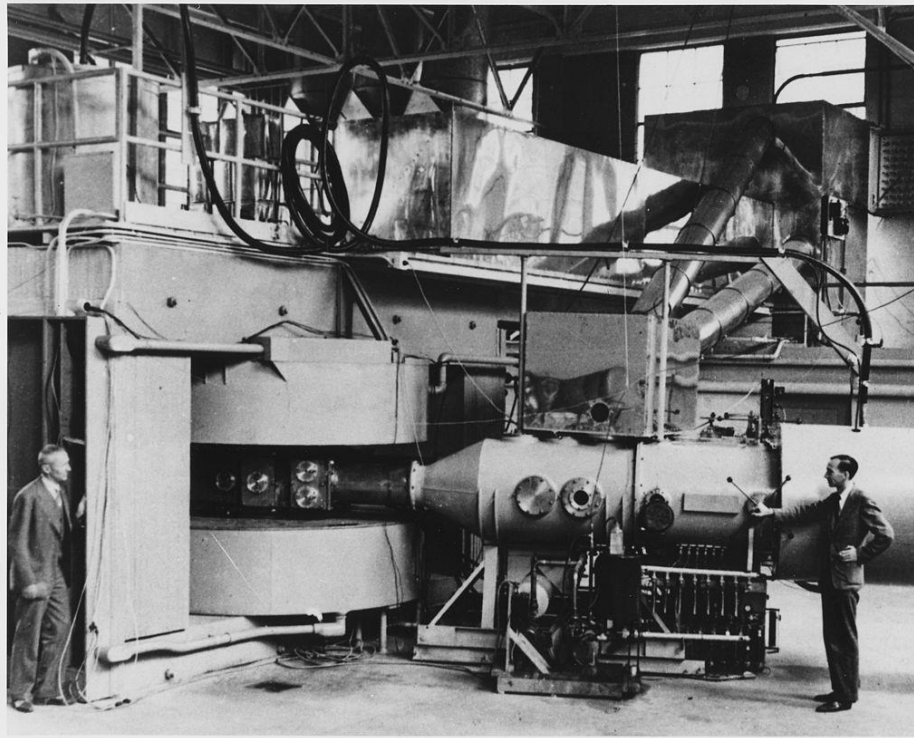
$$QvB = m \frac{v^2}{R}, \text{ or } p = QBR$$

Here ‘ m ’ is the particle's mass and $p = mv$ is its momentum.

A simple experimental technique for finding the momentum of a particle: send it through a region of known magnetic field, and measure the radius of its circular trajectory. This is in fact the standard means for determining the momenta of elementary particles.

Lawrence's 60-inch cyclotron, with magnet poles 60 inches (5 feet, 1.5 meters) in diameter, at the University of California Lawrence Radiation Laboratory, Berkeley, in August, 1939

A particle being accelerated in a cyclotron, and being ejected through a Beamline

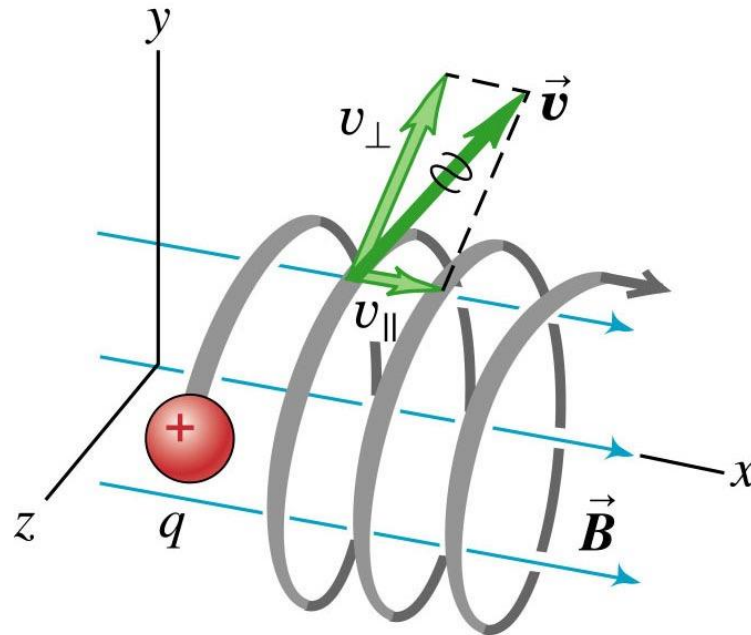


Nobel Prize in Physics (1939)



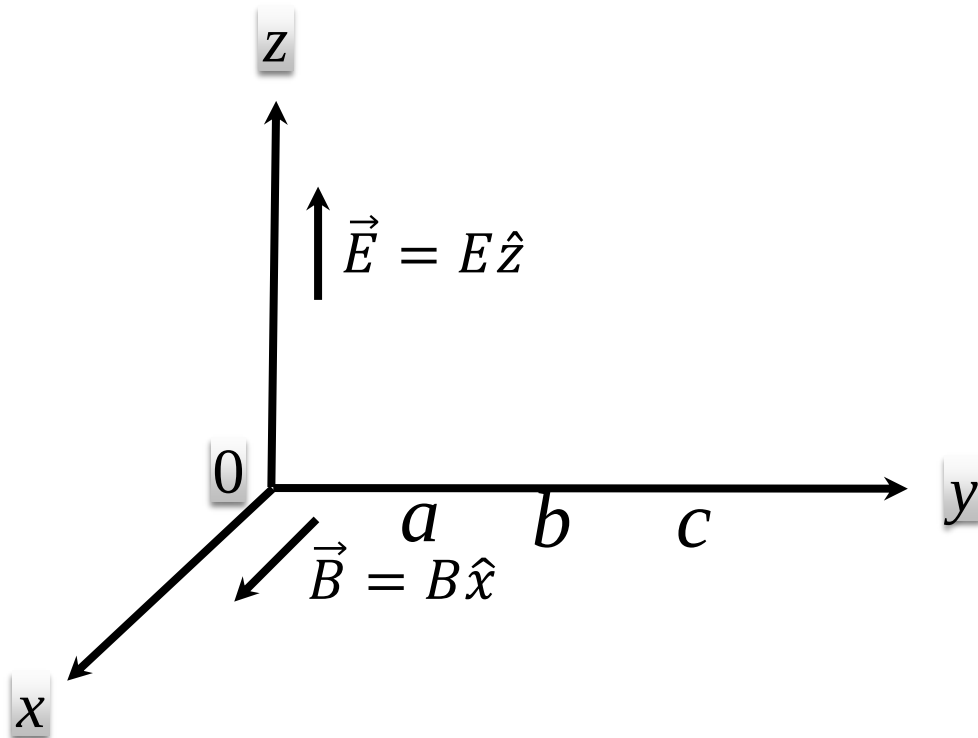
Ernest Lawrence

The charge moves with additional speed ' $v_{\parallel}$ ' parallel to $\vec{B}$, this component of the motion is unaffected by the magnetic field, and the particle moves in a helix



The perpendicular component, $v_{\perp}$, will be the one that determines the radius of the helix

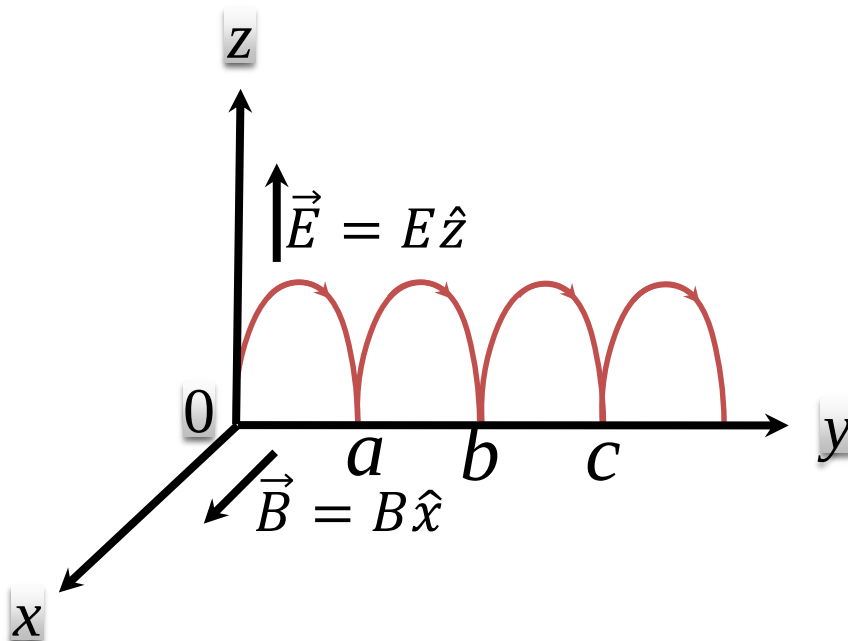
Example: A more exotic trajectory occurs if we include a uniform electric field, at right angles to the magnetic one. Suppose, for instance, that $\vec{B}$ points in the x-direction, and $\vec{E}$ in the z-direction, as shown in Figure. A particle at rest is released from the origin; what path will it follow?



$$\text{Let, } \vec{E} = E\hat{z} \quad \vec{B} = B\hat{x}$$

Initially a positive charged particle is at rest . And this positive charge is released from the origin ($t = 0$, rest)

The particle is at rest , so magnetic force is zero . The electric field accelerates the charge in the $\hat{z}$ direction . As it picks up the speed , a magnetic force develops which pulls the charge to the right means in the $\hat{y}$ direction . As it goes faster , magnetic force becomes stronger ,this force curves the particle back around towards y-axis . Now charged particle moving against the electric force , so it begins to slow down .



The magnetic force then decreases, and the electrical force takes over, bringing the charge to rest at point a . And repeat the whole thing again and again .

Firstly there is no magnetic force in the x-direction .

So position of the charged particle at any time t is $(0, y(t), z(t))$ in the yz -plane and velocity $\vec{v} = (0, \dot{y}, \dot{z})$

and

$$\vec{v} \times \vec{B} = \hat{y}B\dot{z} - \hat{z}B\dot{y}$$

Total force on the charged particle ,

$$\vec{F} = Q(\vec{E} + \vec{V} \times \vec{B}) \quad (1)$$

$$\text{Or , } m\ddot{\vec{a}} = Q (E\hat{z} + \hat{y}B\dot{z} - \hat{z}B\dot{y})$$

$$\text{Or, } m(\ddot{y}\hat{y} + \ddot{z}\hat{z}) = Q (E\hat{z} + \hat{y}B\dot{z} - \hat{z}B\dot{y}) \quad (2)$$

Comparing, both side of $\hat{y}$, $\hat{z}$ in equation (2)

Comparing, both side of $\hat{y}$, $\hat{z}$ in equation (2)

$$m\ddot{y} = QB\dot{z}$$

$$\ddot{y} = (QB/m)\dot{z}$$

$$\ddot{y} = \omega\dot{z} \quad (3)$$

Where $\omega = \frac{QB}{m} =$
cyclotron frequency

$$\text{And , } m\ddot{z} = Q(E - B\dot{y})$$

$$\text{or , } \ddot{z} = (QB/m)\left(\frac{E}{B} - \dot{y}\right)$$

$$\text{or , } \ddot{z} = \omega\left(\frac{E}{B} - \dot{y}\right) \quad (4)$$

Now we have to solve equation (3), (4)

Again differentiate equation (3) , $\ddot{y} = \omega \dot{z}$ (5)

Substituting this into equation (4) :

$$\dot{z} = \omega \left(\frac{E}{B} - \dot{y} \right)$$

$$\text{or, } \ddot{y} = \omega^2 \left(\frac{E}{B} - \dot{y} \right) \quad (6)$$

Solution of equation (6)

$$Y(t) = C_1 \cos \omega t + C_2 \sin \omega t + \frac{Et}{B} + C_3 \quad (7)$$

Substituting this value of eq.7 into equation (3),

$$\ddot{y} = \omega \dot{z}$$

$$[-\omega^2 C_1 \cos \omega t - \omega^2 C_2 \sin \omega t] = \omega \dot{z}$$

Integrating , $Z(t) = C_2 \cos \omega t - C_1 \sin \omega t + C_4$ (8)

Now we apply the boundary condition :

At $t = 0$, particle situated at the origin, and no velocity ,

Boundary condition (i)

$$Y(t=0) = 0$$

$$Z(t=0) = 0$$

Boundary condition (ii)

$$\dot{y} (t=0) = 0$$

$$\dot{z} (t=0) = 0$$

Apply boundary condition (i) ,

$$C_1 + C_3 = 0 \implies C_1 = -C_2$$

From equation (7)

$$C_2 + C_4 = 0 \implies C_2 = -C_4$$

From equation (8)

Apply boundary condition (ii)

Velocity

$$\dot{y} = -C_1 \omega \sin \omega t + C_2 \omega \cos \omega t + E/B \quad (9)$$

$$\dot{z} = -C_2 \omega \sin \omega t - C_1 \omega \cos \omega t \quad (10)$$

$$C_2 \omega + \frac{E}{B} = 0 \implies C_2 = -E/B\omega \quad \text{From equation (9) for } t = 0$$

$$C_1 = 0 \implies C_3 = 0 \quad \text{From equation (10) for } t = 0, v=0$$

Plug all the value of constant into equation (3) & equation (4)

$$Y(t) = \frac{E}{\omega B} (\omega t - \sin \omega t)$$

$$Z(t) = \frac{E}{\omega B} (1 - \cos \omega t)$$

Where , $\frac{E}{\omega B} = R$ (*unit of radius*)

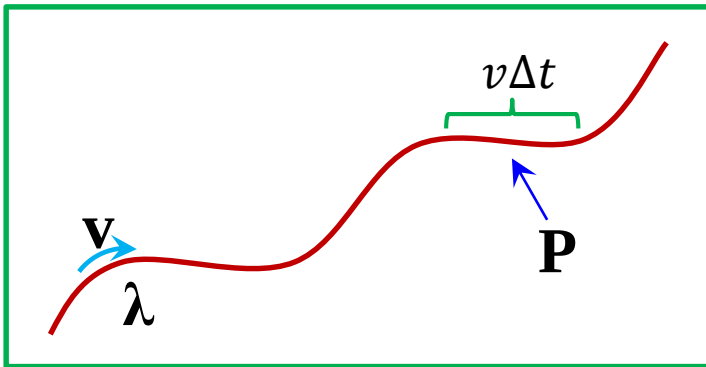
$$(Y - \omega R t)^2 + (Z - R)^2 = (R)^2 \quad (11)$$

This is the formula for a circle of radius R whose center at $(0, \omega R t, R)$ travels in the y-direction at a constant speed; $v = \omega R = \frac{E}{B}$

Currents

- Current in a wire is the charge per unit time passing a given point. By convention, the direction of the current is along the direction of movement of positive charges and opposite to the direction of movement of negative charges.
- It is often due to the movement of negatively charged electrons that contribute current opposite to the movement of electrons. To avoid complication, it is often taken as the positively charged particles that move, so that current direction can be taken as same as those of positive charges.
- Current is measured in the unit of Coulombs per sec or **amperes** (A): $1 \text{ A} = 1 \text{ C/s}$.
- A line charge λ travelling down a wire at speed v as shown in figure constitutes a current, **I**

$$\mathbf{I} = \frac{\Delta Q}{\Delta t} = \frac{\lambda \Delta l}{\Delta t} = \lambda \mathbf{v} \quad \text{--- (1.4)}$$



Here a segment of length $\Delta l = v \Delta t$ carrying a charge $\Delta Q = \lambda v \Delta t$, passes a point $\mathbf{P}$ in a time interval Δt . Current is vector.

- The magnetic force on a segment of current carrying wire as per Lorentz law is,

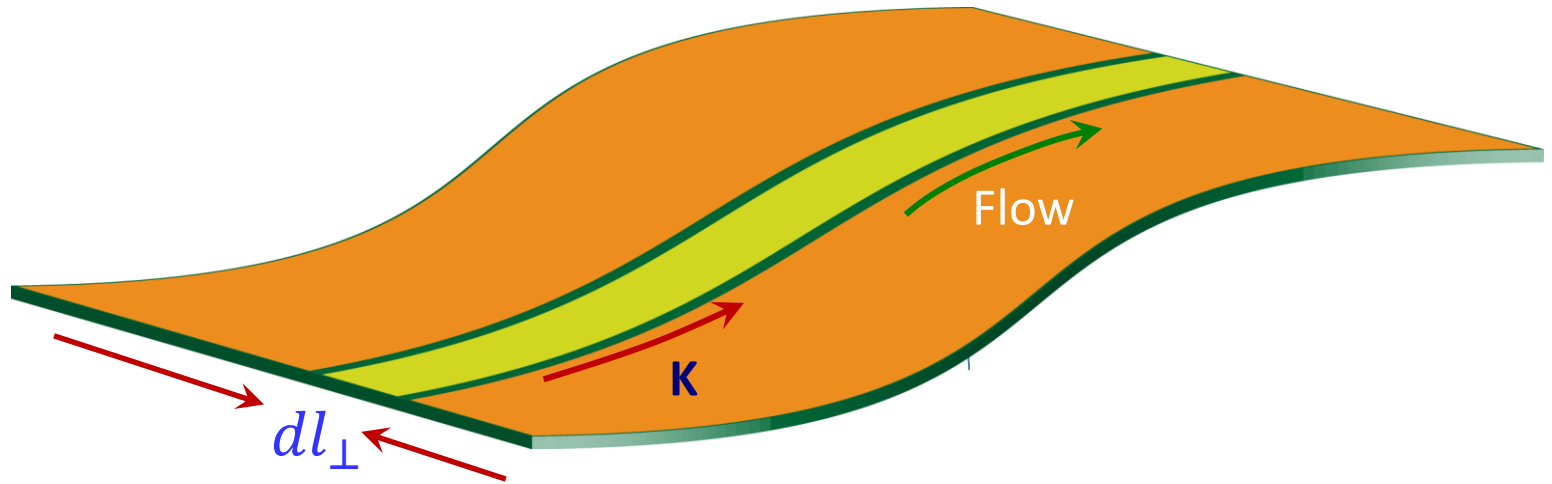
$$\mathbf{F}_{mag} = \int (\mathbf{v} \times \mathbf{B}) dq = \int (\mathbf{v} \times \mathbf{B}) \lambda dl = \int (\mathbf{I} \times \mathbf{B}) dl \quad -- (1.5)$$

- Since $\mathbf{I}$ and $d\mathbf{l}$ both point along the same direction, eq. (1.5) can be written as,

$$\mathbf{F}_{mag} = \int I (d\mathbf{l} \times \mathbf{B}) = I \int (d\mathbf{l} \times \mathbf{B}) \quad -- (1.6)$$

- Since the magnitude of the current passing through the wire is generally constant, it is taken outside the integral.

Surface Current Density



$$\mathbf{K} = \frac{d\mathbf{I}}{dl_{\perp}}$$

If the current in the ribbon is $d\mathbf{I}$

Surface Current Density

- When charges flow over a surface, we describe it by the surface current density, **K**.
- A current carrying surface is shown in Fig. 1.6. Let us consider a ribbon of infinitesimal width $dl_{\perp}$ running along the current flow. If the current in the ribbon is **dI**, the surface current density is,

$$\mathbf{K} = \frac{d\mathbf{I}}{dl_{\perp}} \quad \text{-- (1.7)}$$

- The linear current $d\mathbf{I}$ can be determined from surface charge density σ and surface element da as follows,

$$d\mathbf{I} = \frac{dQ}{dt} = \frac{\sigma da}{dt} = \frac{\sigma(dl_{\parallel})(dl_{\perp})}{dt} = \sigma v(dl_{\perp}) \quad \text{-- (1.8)}$$

- Here $dl_{\parallel}$ represents small length element along the direction of $d\mathbf{I}$, $\mathbf{v}$ is the velocity of charger carriers. By substituting eq. (1.8) into eq.(1.7) gives rise to,

$$\mathbf{K} = \sigma \mathbf{v} \quad \text{-- (1.9)}$$

- In general, $\mathbf{K}$ will vary from point to point over the surface, reflecting the variations in σ and $\mathbf{v}$.
- The magnetic force on the surface current is,

$$\mathbf{F}_{mag} = \int (\mathbf{v} \times \mathbf{B}) \sigma da = \int (\mathbf{K} \times \mathbf{B}) da \quad -- (1.10)$$

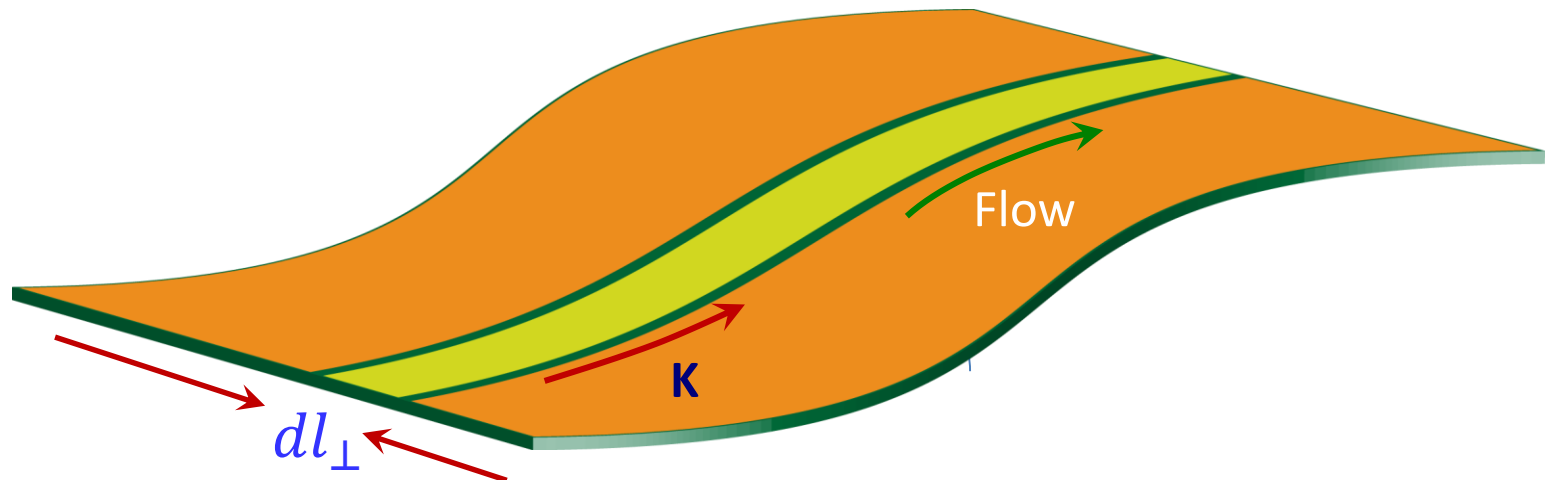
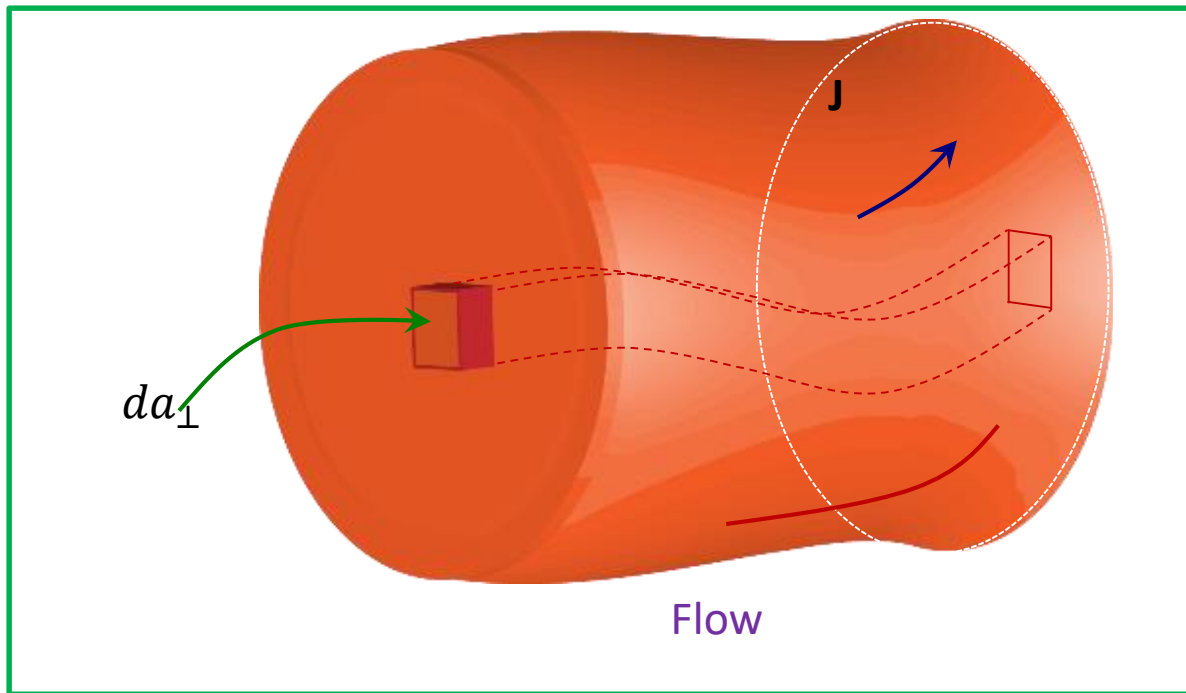


Fig. 1.6

Volume Current Density

$$\mathbf{J} = \rho \mathbf{v}$$

$$\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}}$$



Volume Current Density:

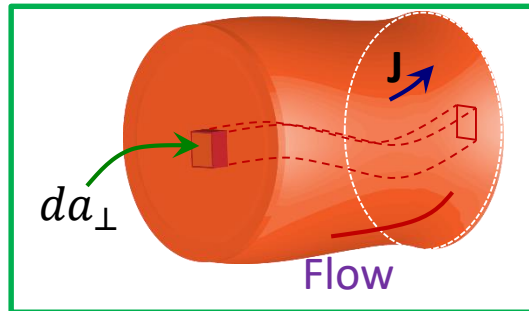
- When the flow of charge is distributed throughout a three dimensional region, it is described as volume current density, $\mathbf{J}$. It is defined as,

$$\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}} \quad \text{-- (1.11)}$$

- Here $d\mathbf{I}$ is the current flowing through a tube of infinitesimal cross sectional area $da_{\perp}$ as shown in below figure.
- $\mathbf{J}$ is the current per unit area perpendicular to the flow. $\mathbf{J} = \rho \mathbf{v}$ -- (1.12)
- Similar to the case of surface current density, if we take ρ as the volume charge density and $\mathbf{v}$ as the velocity of charge carriers, then

$$\mathbf{F}_{mag} = \int (\mathbf{v} \times \mathbf{B}) \rho d\tau = \int (\mathbf{J} \times \mathbf{B}) d\tau \quad \text{-- (1.13)}$$

- The magnetic force on the volume current density is therefore,



- According to eq. (1.11), the current crossing a surface S can be written as,

$$\mathbf{I} = \int_S \mathbf{J} d\mathbf{a}_\perp = \int_S \mathbf{J} \cdot d\mathbf{a} \quad \text{--- (1.14)}$$

- The total charge per unit time leaving a volume V, is

$$\oint_S \mathbf{J} \cdot d\mathbf{a} = \int_V \nabla \cdot \mathbf{J} d\tau \quad \text{--- (1.15)}$$

$$\int_V \nabla \cdot \mathbf{J} d\tau = -\frac{d}{dt} \int_V \rho d\tau = -\int_V \frac{d\rho}{dt} d\tau \quad \text{-- (1.16)}$$

- Because, the charge is conserved, whatever flows out through the surface must be at the expense of remaining inside. So,

$$\boxed{\nabla \cdot \mathbf{J} = -\frac{d\rho}{dt}} \quad \text{-- (1.17)}$$

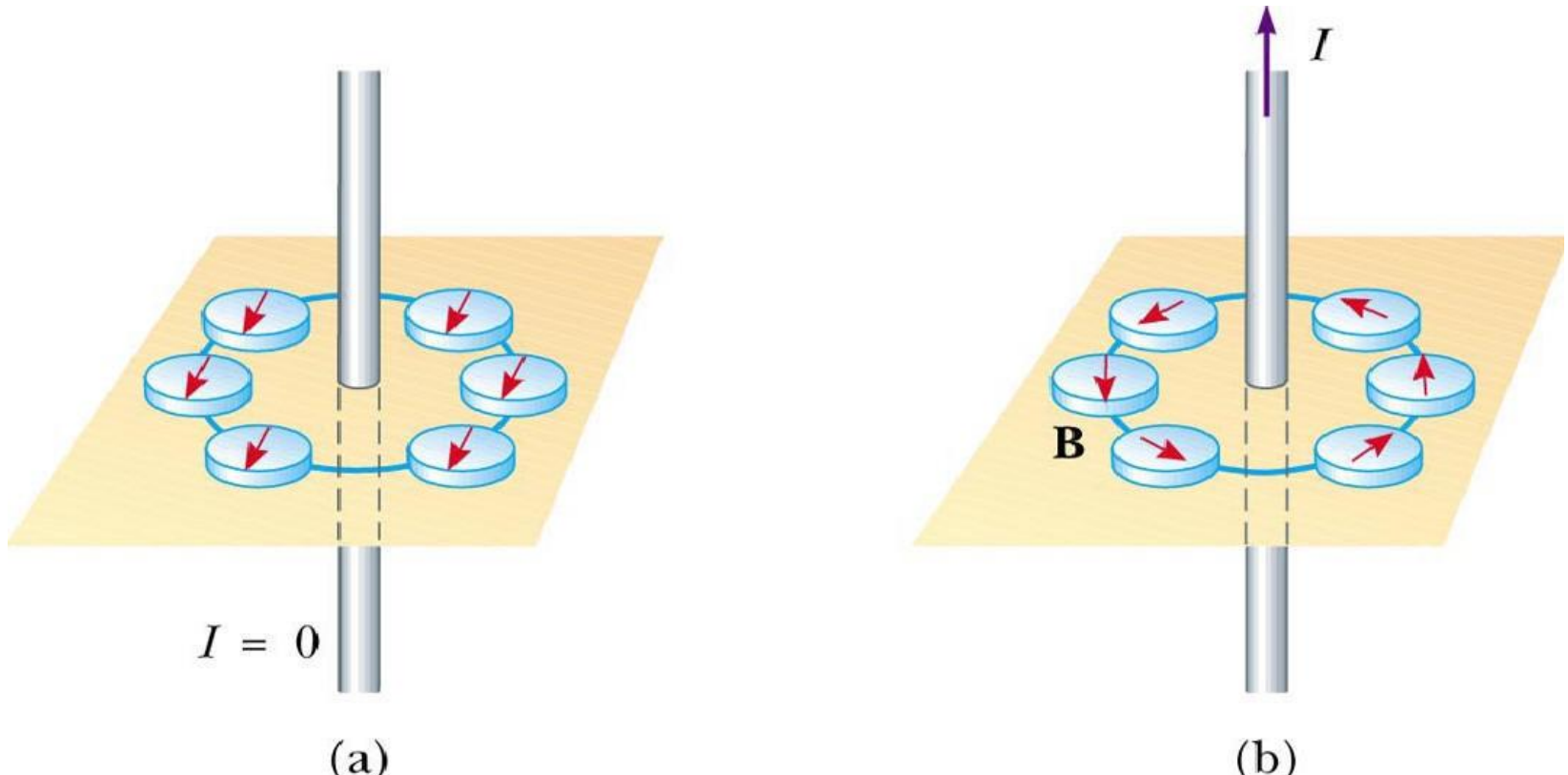
- Since, it is applied to any volume, it is concluded that,
- Eqn. 1.17 is called the continuity equation.

Biot-Savart's Law

A wire with current behaves like a magnet.

A simple experiment demonstrates that a current-carrying conductor produces a magnetic field. In this experiment several compass needles are placed in a horizontal plane near a long vertical wire.

When $I = 0$ all compass needles point to the North Pole of Earth. However, when the wire carries a strong current I , all the needles deflect in directions tangent to the circle.



Steady currents

Stationary charges → constant electric fields: Electrostatics.

Steady currents' → constant magnetic fields: Magnetostatics.

Steady current means that a continuous flow that goes on forever without change and without charge piling up anywhere. They represent *suitable approximations* as long as the fluctuations are reasonably slow. $\nabla \cdot \mathbf{J} = 0$

Biot-Savart Law

The Biot-Savart Law relates magnetic fields to the currents which are their sources. In a similar manner, Coulomb's law relates electric fields to the point charges which are their sources.

It is named after **Jean-Baptiste Biot** and **Félix Savart** who discovered this relationship in 1820

The Biot-Savart law for Line current, Surface current and Volume current.

Line current

$$B = \frac{\mu_o}{4\pi} \int \frac{i \, dl \times \hat{r}}{r^2}$$

Surface current

$$B = \frac{\mu_o}{4\pi} \int \frac{\vec{K} ds \times \hat{r}}{r^2}$$

Volume current

$$B = \frac{\mu_o}{4\pi} \int \frac{\vec{J} dV \times \hat{r}}{r^2}$$

The Biot-Savart law for Line current, Surface current and Volume current

❑ Biot-Savart law for Line current

$$B = \frac{\mu_o}{4\pi} \int \frac{i \, dl \times \hat{r}}{r^2}$$

❑ Biot-Savart law for Surface current

$$B = \frac{\mu_o}{4\pi} \int \frac{\vec{K} \, ds \times \hat{r}}{r^2}$$

❑ Biot-Savart law for Volume current

$$B = \frac{\mu_o}{4\pi} \int \frac{\vec{J} \, dV \times \hat{r}}{r^2}$$

Applications of Biot-Savart Law

The Biot-Savart law describes the magnitude of the magnetic field at distance 'S' surrounding an infinitely long current-carrying wire.

The direction of the magnetic field surrounding an infinitely long current-carrying wire is obtained by the right-hand rule.

Thumb = **direction of current I**, fingers = **direction of B**.

The Force is perpendicular to the palm.

Magnetic field due to an infinitely long current-carrying wire



$$B = \frac{\mu_0 I}{2\pi s}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

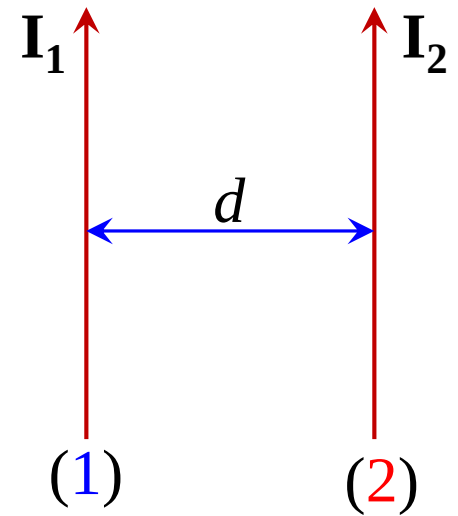
$$\text{N A}^{-2}$$

As an application, let's find the force of attraction between two long, parallel wires a distance 'd' apart, carrying currents I_1 and I_2

The field at (2) due to (1) is

points into the page

$$B = \frac{\mu_0 I_1}{2\pi d}$$

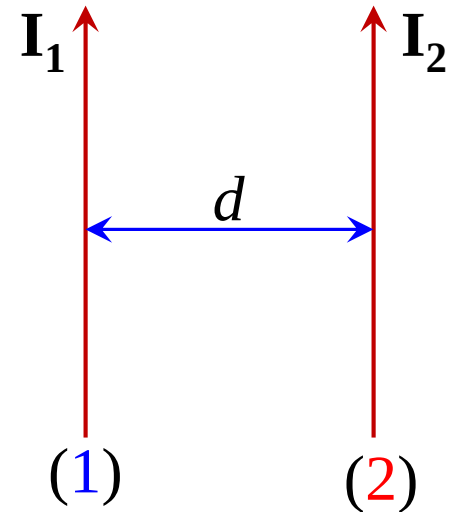


The field at (2) due to (1) is

$$B = \frac{\mu_0 I_1}{2\pi d} \longrightarrow \text{points into the page}$$

The Lorentz force law (in the form appropriate to line currents given by below equation,

$$\mathbf{F}_{mag} = I \int (\overrightarrow{dl} \times \vec{B})$$



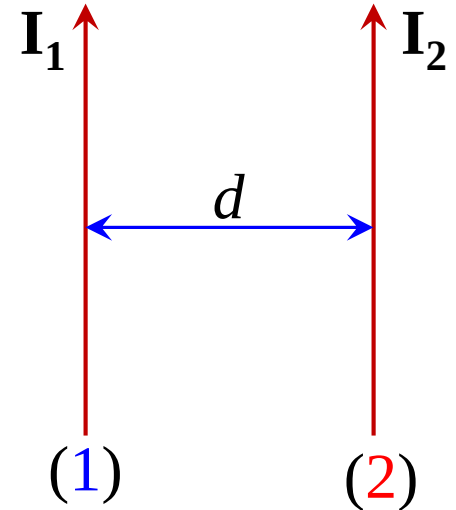
This equation predicts a force directed towards (1), of magnitude

$$F = I_2 \left(\frac{\mu_0 I_1}{2\pi d} \right) \int dl$$

The field at (2) due to (1) is

The Lorentz force law (in the form appropriate to line currents given by below equation,

$$\mathbf{F}_{mag} = I \int (\overrightarrow{dl} \times \vec{B})$$



This equation predicts a force directed towards (1), of magnitude



$$F = I_2 \left(\frac{\mu_0 I_1}{2\pi d} \right) \int dl$$

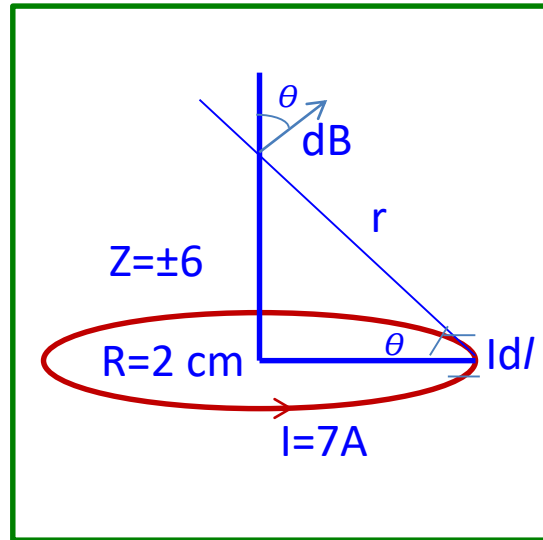
The total force, not surprisingly, is infinite, but the force per unit length is

$$f = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d}$$

Few examples on Biot-Savart law

Example: A circular loop located on $x^2+y^2=4^2$, ' $z=0$ ' carries a direct current of 7 A along $\hat{\phi}$. Determine the magnetic flux density B at $(0,0,6\text{ cm})$ and $(0,0,-6\text{ cm})$.

Solution: Consider the circular loop shown in the adjacent figure.



The magnetic field

$$B = \text{????? Newton}/(\text{A} \cdot \text{m}) \text{ (or Tesla = Newton}/(\text{A} \cdot \text{m}))$$

General Method

Example-1: Determine the magnetic field (B) at distance 'z' above the center of a circular loop of radius 'R' which carries a steady current 'I'.

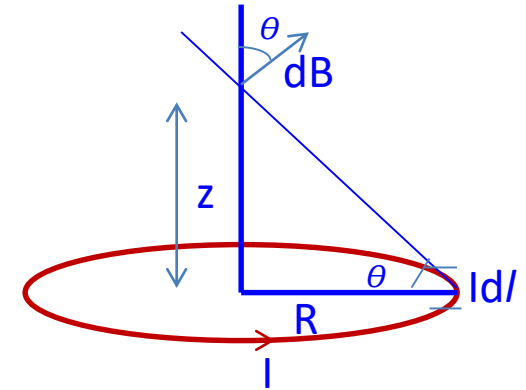
$$B = \frac{\mu_o}{4\pi} \int \frac{i \, dl \times \hat{r}}{r^2}$$

$$B = \frac{I \mu_o}{4\pi} \int \frac{dl \cos(\theta)}{r^2}$$

$$= \frac{I \mu_o \cos(\theta)}{4\pi r^2} \int dl$$

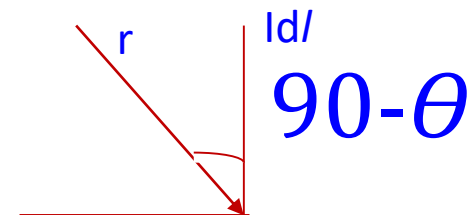
$$B = \frac{I \mu_o}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$$

$$dB/dz = -\frac{3Iz \mu_o}{2} \frac{R^2}{(R^2 + z^2)^{5/2}}$$



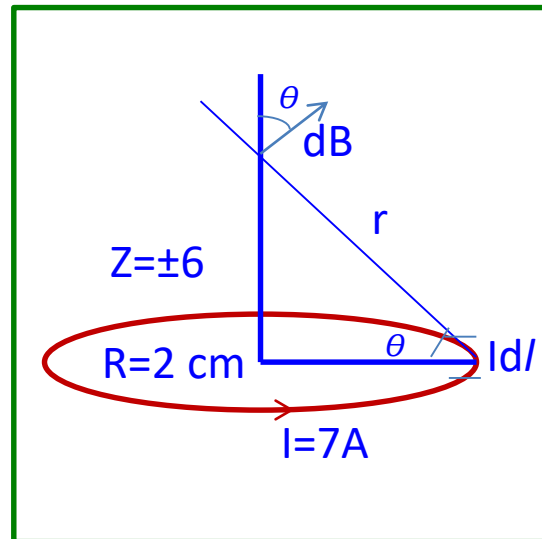
$$\int dl = 2\pi R$$

$$\cos(\theta) = R/r$$



Example-2: A circular loop located on $x^2+y^2= 4^2$, ' $z=0$ ' carries a direct current of 7 A along $\hat{\phi}$. Determine the magnetic flux density B at (0,0,6 cm) and (0,0,-6 cm).

Solution: Consider the circular loop shown in the adjacent figure.

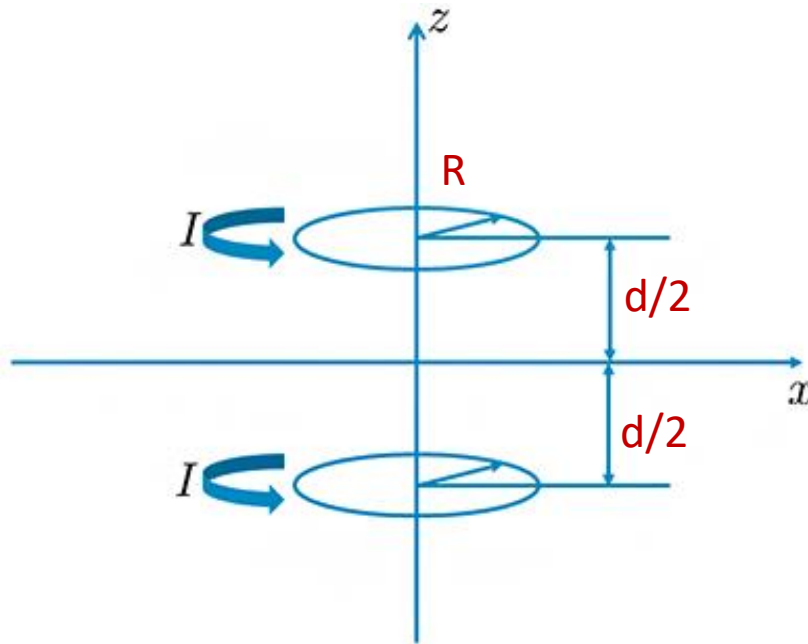


The magnetic field

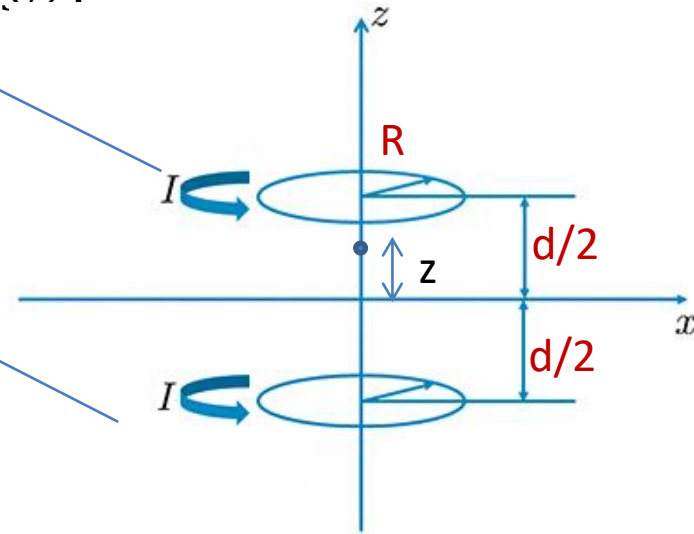
$$B(z) = \frac{I \mu_o}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$$

Substituting $I = 7 \text{ A}$, $R = 2 \text{ cm}$, $z = +6 \text{ cm}$, $\mu_o = 4\pi \times 10^{-7} \text{ N A}^{-2}$,
 $B = 6.95 \times 10^{-6} \text{ Newton/(A. m)}$ (or Tesla = Newton/(A. m))

A magnetic field on the axis of a circular current loop is not very uniform and a more uniform field on the axis can be produced by using two-such loops a distance 'd' apart as shown below



$$B = \frac{\mu_o I R^2}{2} \left[\frac{1}{(R^2 + (\frac{d}{2} + z)^2)^{3/2}} + \frac{1}{(R^2 + (\frac{d}{2} - z)^2)^{3/2}} \right]$$

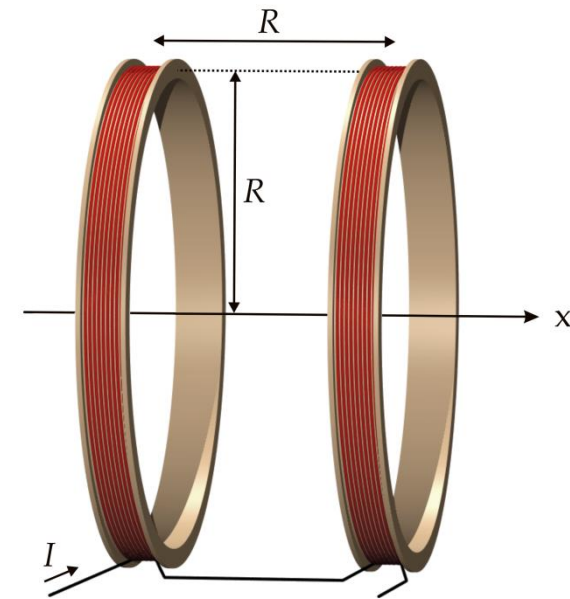
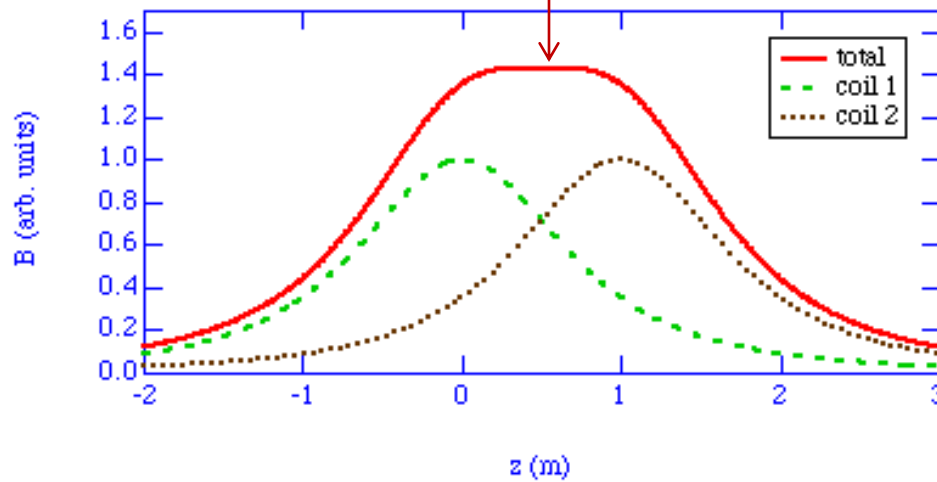


Helmholtz coils

$$\frac{\partial B}{\partial z} = \frac{3\mu_o I R^2}{2} \left[\frac{-(d/2 + z)}{(R^2 + (\frac{d}{2} + z)^2)^{5/2}} + \frac{(\frac{d}{2} - z)}{(R^2 + (\frac{d}{2} - z)^2)^{5/2}} \right]$$

$$\frac{\partial B}{\partial z} = 0 \Rightarrow \text{at } z = 0 \text{ i.e. at the point midway between the two-loops}$$

Zone of constant magnetic field



The second derivative of 'B' can also be made to zero at this point such an arrangement is known as Helmholtz coil

$$\frac{\partial^2 B}{\partial z^2} = 3\mu_o I R^2 \left[\frac{(d^2 - R^2)}{(R^2 + \frac{d^2}{4})^{7/2}} \right]$$

$$\frac{\partial^2 B}{\partial z^2} = 0 \text{ when } d = R$$

Condition for Helmholtz coil

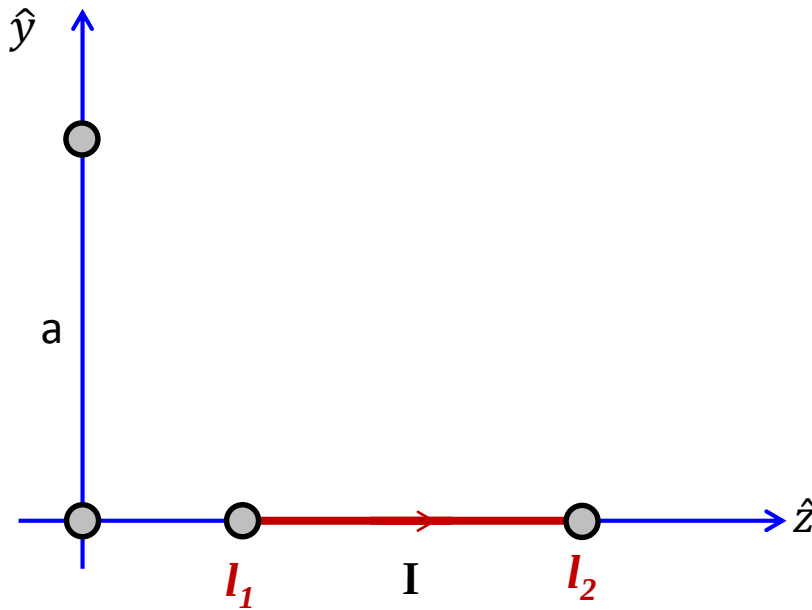
at $z = 0$

$$B = \frac{\mu_o I R^2}{2} \left[\frac{1}{(R^2 + (\frac{R}{2})^2)^{3/2}} + \frac{1}{(R^2 + (\frac{R}{2})^2)^{3/2}} \right] = \mu_o I R^2 \frac{1}{(5R^2/4)^{3/2}} = \frac{8\mu_o I}{R5^{3/2}}$$

Few EXAMPLES

Example: The Macroscopic Magnetic Field due to a Steady Current Flowing in a Finite Length Wire.

What is the magnetic field $\mathbf{B}$ associated with a steady current $\mathbf{I}$ flowing down a finite length filamentary wire of length l ?



Biot-Savart law

$$\vec{B}(r) = \frac{\mu_0}{4\pi} \int \frac{I \vec{dl} \times \hat{r}}{r^2}$$

$$dl \times \hat{r} = |dl| |\hat{r}| \sin(90 + \theta)$$

$$dl \times \hat{r} = |dl| |\hat{r}| \cos \theta$$

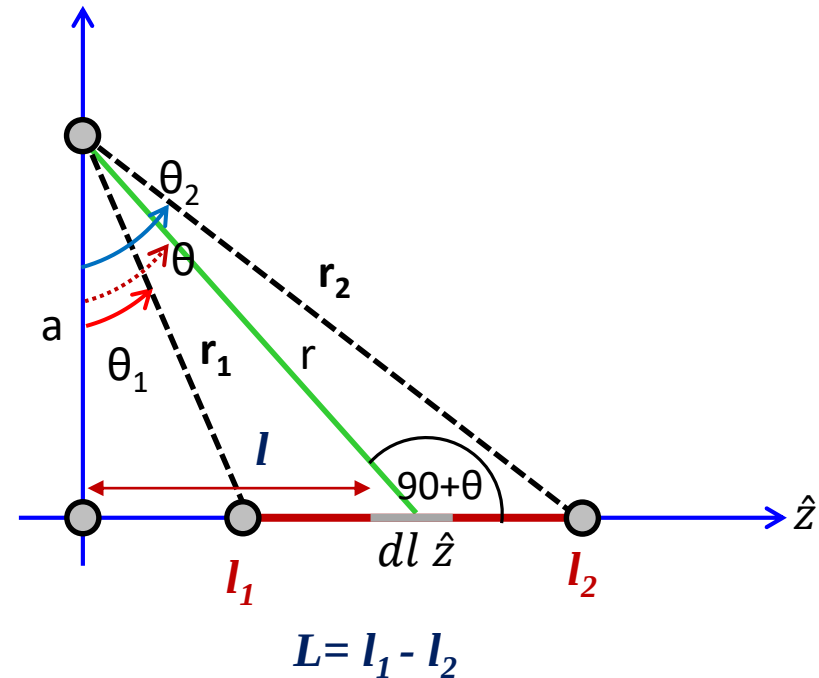
$$|\hat{r}| = 1$$

$$l = a \tan \theta$$

$$dl = a \sec^2 \theta d\theta$$

$$dl \times \hat{r} = a \sec^2 \theta d\theta \cos \theta$$

Consider a small element $dl \hat{z}$



$$r = \sqrt{a^2 + l^2}$$

$$\vec{B}(r) = \frac{\mu_0 I}{4\pi} \int_{\theta=\theta_1}^{\theta=\theta_2} \frac{a \sec^2 \theta \cos \theta d\theta}{a^2 + l^2} \hat{\phi}$$

$$l = a \tan \theta$$

$$\vec{B}(r) = \frac{\mu_0 I}{4\pi a^2} \int_{\theta=\theta_1}^{\theta=\theta_2} \frac{a \sec^2 \theta \cos \theta d\theta}{1 + \tan^2 \theta} \hat{\phi}$$

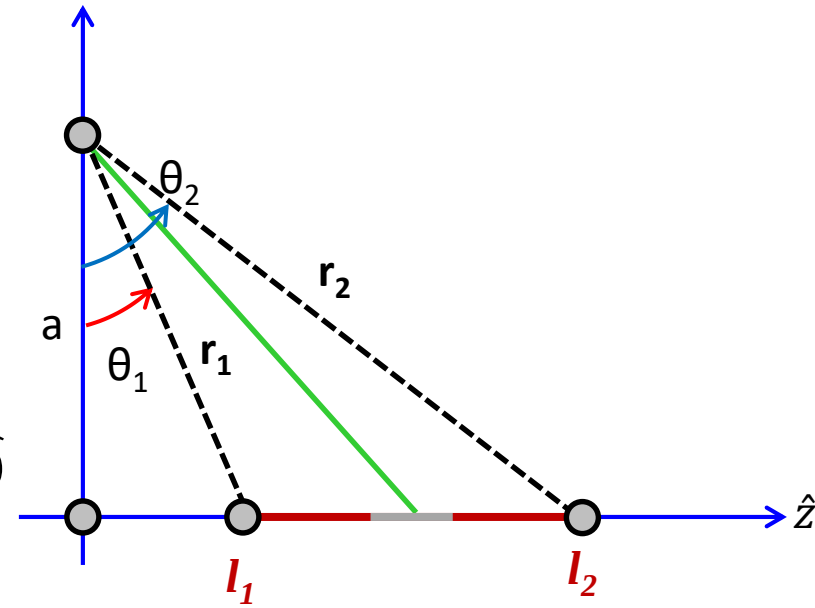
$$\vec{B}(r) = \frac{\mu_0 I}{4\pi a} \int_{\theta=\theta_1}^{\theta=\theta_2} \cos \theta d\theta \hat{\phi}$$

$$\vec{B}(r) = \frac{\mu_0 I}{4\pi a} [\sin \theta_2 - \sin \theta_1]$$

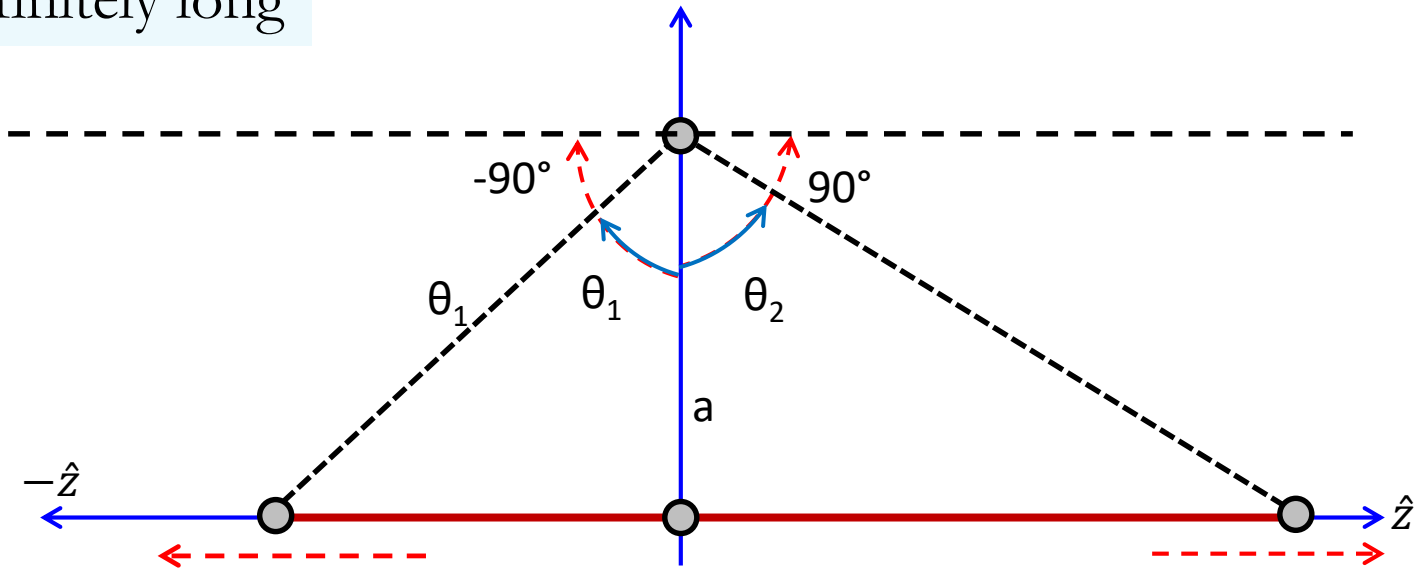
Here

$$\sin \theta_1 = \frac{l_1}{\sqrt{a^2 + l_1^2}}$$

$$\sin \theta_2 = \frac{l_2}{\sqrt{a^2 + l_2^2}}$$



If this wire is infinitely long



$$\vec{B}(r) = \frac{\mu_0 I}{4\pi a} [\sin \theta_2 - \sin \theta_1]$$

$$\sin \theta_2 \rightarrow 90^\circ$$

$$\sin \theta_1 \rightarrow -90^\circ$$

$$\vec{B}(r) = \frac{\mu_0 I}{4\pi a} [1 - (-1)]$$

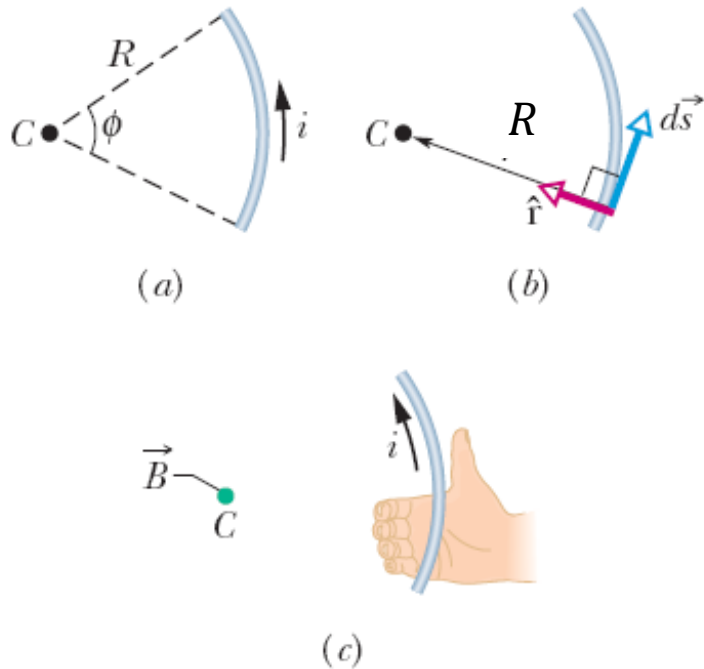
$$\sin 90^\circ = 1$$

$$\vec{B}(r) = \frac{\mu_0 I}{2\pi a}$$

$$\sin(-90^\circ) = -1$$

Here B is inversely proportional to the distance ' a ' from the wire and is comparable to the electric field produced by infinite line charge density. Direction is along $\hat{\phi}$

Magnetic Field due to a Current in a Circular Arc of Wire



The right-hand rule reveals the field's direction at the center.

$$B = \frac{\mu_0 i (2\pi)}{4\pi R} = \frac{\mu_0 i}{2R}$$

At center of full circle

$$\vec{B}(r) = \frac{\mu_0}{4\pi} \int \frac{I \vec{ds} \times \hat{r}}{r^2}$$

$$dB = \frac{\mu_0 i ds \sin 90^\circ}{4\pi R^2}$$

$$= \frac{\mu_0 i ds}{4\pi R^2}$$

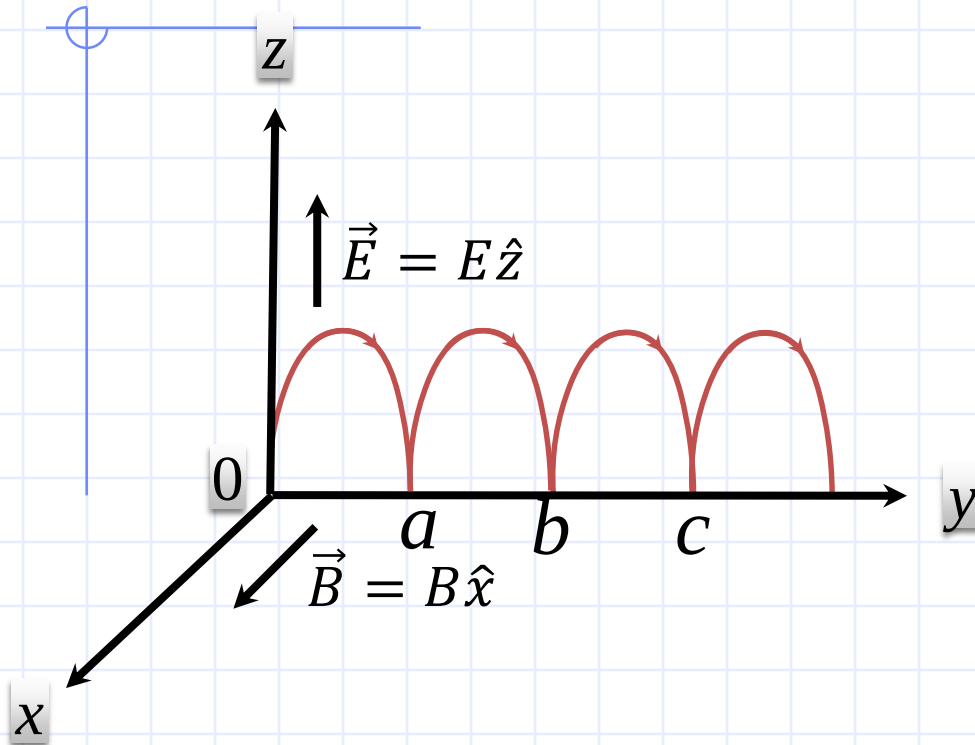
$$B = \int dB = \frac{\mu_0}{4\pi} \int_0^\phi \frac{i R d\phi}{R^2}$$

$$= \frac{\mu_0 i}{4\pi R} \int_0^\phi d\phi$$

$$B = \frac{\mu_0 i \phi}{4\pi R}$$

At center of circular arc

GEx.5.2: A more exotic trajectory occurs if we include a uniform electric field, at right angles to the magnetic one. Suppose, for instance, that $\vec{B}$ points in the x -direction, and $\vec{E}$ in the z -direction, as shown in Figure. A particle at rest is released from the origin; what path will it follow?



$$v(t=0) = 0 \quad v(t=\Delta t) = v_z = qE\Delta t$$

$$\overrightarrow{F_{mag}} = qv_z B \hat{y}$$

$$m \frac{d\vec{v}}{dt} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$m \frac{d^2 y}{dt^2} = q \frac{dz}{dt} B$$

$$m \frac{d^2 z}{dt^2} = q(E - \frac{dy}{dt} B)$$

$$\dot{v}_y = \omega_c v_z$$

$$\dot{v}_z = \omega_c \left(\frac{E}{B} - v_y \right)$$

Assumption

$$v_y(t) = V_y(t) + V_{yc} \quad V_{yc} = \text{const} \quad v_z(t) = V_z(t)$$

$$\begin{aligned} \dot{V}_y &= \omega_c V_z(t) & V_{yc} &= \frac{E}{B} \\ \dot{V}_z &= \omega_c \left(\frac{E}{B} - V_y(t) - V_{yc} \right) \end{aligned}$$

Modified equations

$$\begin{aligned} \dot{V}_y &= \omega_c V_z & V_y &= C_1 \sin \omega_c t + C_2 \cos \omega_c t \\ \dot{V}_z &= -\omega_c V_y & V_z &= -C_2 \sin \omega_c t + C_1 \cos \omega_c t \end{aligned}$$

Initial condition

$$\begin{aligned} v_y(t=0) = 0 &= V_y(t=0) + V_{yc} \Rightarrow C_2 = -\frac{E}{B} \\ v_z(t=0) = 0 &= V_z(t=0) \Rightarrow C_1 = 0 \end{aligned}$$

Velocity Equation

$$v_y = \frac{E}{B} (1 - \cos \omega_c t) \quad v_z = \frac{E}{B} \sin \omega_c t$$

Velocity Equation

$$v_y = \frac{E}{B}(1 - \cos\omega_c t) \quad v_y = \frac{E}{B}\sin\omega_c t$$

Equation of motion

$$y = \frac{E}{\omega_c B}(\omega_c t - \sin\omega_c t) + C_3 \quad z = -\frac{E}{\omega_c B}\cos\omega_c t + C_4$$

Initial condition

$$y(0)=z(0)=0$$

$$\begin{aligned} y(t=0) &= 0 & C_3 &= 0 \\ z(t=0) &= 0 & C_4 &= \frac{E}{\omega_c B} \end{aligned}$$

$$y = \frac{E}{\omega_c B}(\omega_c t - \sin\omega_c t) = R(\omega_c t - \sin\omega_c t) = \bar{y} + R\omega_c t$$

$$z = \frac{E}{\omega_c B}(1 - \cos\omega_c t) = R(1 - \cos\omega_c t) = \bar{z} + R$$

$$\bar{y} = -R\sin\omega_c t$$

$$\bar{z} = -R\cos\omega_c t$$

Cycloid motion

$$(y - R\omega_c t)^2 + (z - R)^2 = R^2$$

Few Exercise Problems

Current is flowing radially from a spherical surface of radius a to concentric spherical surface of radius b ($b > a$). The current density at a distance r ($a < r < b$) is given by

$$\vec{J}(\vec{r}) = J_0 \left(\frac{r^2}{a^2} \right) e^{-kt} \hat{r}$$

Continuity equation

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\vec{J} = J_0 \left(\frac{r^2}{a^2} \right) e^{-kt} \hat{r}.$$

$$\vec{\nabla} \cdot \vec{J} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 J_0 \frac{r^2}{a^2} e^{-kt} \right)$$

$$= \frac{J_0}{a^2} e^{-kt} \frac{1}{r^2} \frac{\partial}{\partial r} (r^4)$$

$$= \frac{4J_0}{a^2} e^{-kt} r$$

$$\rho(r, t) = -\frac{4J_0 r}{a^2} \int e^{-kt} dt$$

$$= \frac{4J_0 r}{ka^2} e^{-kt}$$

(a) A charged particle is moving with velocity $\vec{v} = v\hat{k}$ in a region where $\vec{E} = E\hat{i}$ and $\vec{B} = B\hat{j}$. Show that the Lorentz force is zero if $v = E/B$.

$$\vec{F} = q \left[\vec{E} + \vec{v} \times \vec{B} \right]$$

$$= qE\hat{i} + qv\hat{k} \times B\hat{j}$$

$$= (qE - qvB)\hat{i}$$

clearly $F = 0$ if $v = E/B$.

(b) Show that if a particle moves in a time independent electric field $\vec{E} = -\vec{\nabla} \phi(x, y, z)$ and any magnetic field, then the energy $\frac{1}{2}mv^2 + q\phi$ is a constant.

$$\vec{F} = q \left(\vec{E} + \vec{v} \times \vec{B} \right)$$

$$m \frac{d\vec{v}}{dt} = q \left(\vec{E} + \vec{v} \times \vec{B} \right)$$

$$m\vec{v} \cdot \frac{d\vec{v}}{dt} - q\vec{v} \cdot \vec{E} = q\vec{v} \cdot \left(\vec{v} \times \vec{B} \right) = 0$$

$$\frac{d}{dt} \left[\frac{1}{2} m v^2 \right] - q \vec{v} \cdot \vec{E} = 0$$

$$\vec{E} = -\vec{\nabla} \phi$$

$$\frac{d\phi}{dt} = \frac{d\phi}{dx} \frac{dx}{dt} + \frac{d\phi}{dy} \frac{dy}{dt} + \frac{d\phi}{dz} \frac{dz}{dt} = \vec{\nabla} \phi \cdot \vec{v}$$

Thus we have

$$\frac{d}{dt} \left[\frac{1}{2} m v^2 \right] - q \vec{v} \cdot \vec{E} = 0$$

$$\frac{d}{dt} \left[\frac{1}{2} m v^2 \right] - q \vec{v} \cdot \vec{\nabla} \phi = 0$$

$$\frac{d\phi}{dt} = \frac{d\phi}{dx} \frac{dx}{dt} + \frac{d\phi}{dy} \frac{dy}{dt} + \frac{d\phi}{dz} \frac{dz}{dt} = \vec{\nabla} \phi \cdot \vec{v}$$

Thus we have

$$\frac{d}{dt} \left[\frac{1}{2} m v^2 \right] - q \vec{v} \cdot \vec{E} = 0$$

$$\frac{d}{dt} \left[\frac{1}{2} m v^2 \right] - q \vec{v} \cdot \vec{\nabla} \phi = 0$$

$$\frac{d}{dt} \left[\frac{1}{2} m v^2 \right] - q \frac{d\phi}{dt} = 0$$

$$\frac{d}{dt} \left[\frac{1}{2} m v^2 - q \phi \right] = 0$$

The above expression implies that $\frac{1}{2} m v^2 + q \phi$ is a constant.

A particle with charge q is projected successively along the x - and y -axis with the same speed v . The force on the particle in these situations is given by

$qvB\left(-\frac{1}{2}\hat{j} + \frac{\sqrt{3}}{2}\hat{k}\right)$ & $qvB\left(\frac{1}{2}\hat{i}\right)$ respectively. Find the direction of the magnetic field.

$$\vec{B} = B_1\hat{i} + B_2\hat{j} + B_3\hat{k}$$

$$\vec{F}_1 = qvB \left(-\frac{1}{2}\hat{j} + \frac{\sqrt{3}}{2}\hat{k} \right)$$

$$= qv\hat{i} \times [\hat{i}B_1 + \hat{j}B_2 + \hat{k}B_3]$$

$$= qv \left[\hat{k}B_2 - \hat{j}B_3 \right]$$

The above expression implies that

$$B_2 = \frac{\sqrt{3}}{2}B; \quad B_3 = \frac{1}{2}B$$

Again,

$$\begin{aligned} \vec{F}_2 &= qvB \frac{1}{2}\hat{i} \\ &= qv\hat{j} \times \left[-\hat{i}B_1 + \hat{j}B_2 + \hat{k}B_3 \right] \\ &= qv \left[-\hat{k}B_1 + \hat{i}B_3 \right] \end{aligned}$$

The above expression implies

$$B_1 = 0 \quad B_3 = \frac{B}{2}$$

Therefore, $\vec{B} = \frac{B}{2} \left(\sqrt{3}\hat{j} + \hat{k} \right)$